

sion of remote work, and supply-side shocks to junior labor. We find little support for any of these explanations. The absence of pre-trends and the inclusion of industry-by-seniority-by-time fixed effects suggest that post-COVID sectoral dynamics do not drive our results. Re-estimating our triple-difference specification separately using employment in high- and low-remote-suitability positions yields similar declines in junior employment, suggesting that the results are not driven by shifts in remote work. Finally, job postings decline sharply at adopting firms after 2023, indicating that the reduction in junior employment reflects weaker labor demand rather than constraints on labor supply. Taken together, this evidence suggests that, although these factors are important for broader labor-market trends, they are unlikely to explain the specific effect identified by our empirical design.

We next examine the mechanisms underlying the decline in junior employment. Using our linked employer-employee data, we decompose workforce dynamics into inflows (hires), outflows (separations), and internal promotions. We find that the decline in junior employment at adopting firms is driven primarily by a substantial reduction in hiring. Separation rates for juniors in adopting firms also decreased relative to non-adopters, but the reduction in hiring was considerably larger, leading to a net decline in junior positions. Promotion rates rose by a small but statistically significant amount, though the magnitude is economically small relative to the hiring channel.

Taken together, these results indicate that GenAI adoption is associated with reduced junior employment. However, the relatively early and pronounced declines—starting to emerge soon after the release of ChatGPT—may seem surprising, as automation impacts typically materialize more slowly. This suggests that the decline may not reflect immediate task automation but rather forward-looking adjustments by firms: the rapid diffusion of GenAI may have shifted firms’ expectations, leading them to scale back hiring for roles they predict will be automated in the near future, consistent with broader evidence on asymmetric employment adjustment (Ilut et al., 2018). We formalize this mechanism in a simple dynamic model in which expectations of *future* automation—combined with labor-adjustment costs—lead firms to reduce hiring *today* (Supplemental Appendix A.5). Several patterns are consistent with this interpretation: GenAI adoption accelerated sharply in early 2023; mentions of “AI” in firms’ earnings calls began rising in 2022Q4; prominent U.S. companies publicly adjusted hiring practices in early 2023,

Table 4: Hiring, Separations, and Promotions

	Hiring	Separation	Promotion (rate)
Treat $\times$ Post	−4.006*** (0.221)	−1.089*** (0.163)	0.033** (0.014)
Pre-period mean	5.059	4.154	0.822
Observations	7,941,179	7,941,179	7,690,581
Clusters (firms)		280,866	

Notes: This table reports the estimated β from Equation 15. Estimation begins in 2018 to exclude early-sample trends. Standard errors clustered by firm in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6 Conclusion

This paper provides early, large-scale evidence that the diffusion of GenAI since 2023 is associated with *seniority-biased* employment effects within firms. Using résumé and job-posting data linked to roughly 281,000 U.S. firms, together with a firm-level adoption measure based on “GenAI integrator” vacancies, we document a decline in junior employment relative to senior employment at adopting firms compared with non-adopting firms. This decline is concentrated in high-exposure occupations: within adopting firms, junior employment in high-exposure occupations contracts relative to low-exposure occupations after 2022, while the corresponding senior series continues to rise. Task-level evidence from job postings points to the same mechanism: GenAI-exposed tasks become less likely to appear in junior task bundles at adopting firms. Finally, decomposing employment flows shows that the decline is driven primarily by slower hiring rather than increased separations.

These findings should be interpreted with caution. Adopting firms differ systematically in size, workforce composition, and industry. Although our research designs absorb many observable and unobservable differences across firms, sectors, and time, unobserved confounding factors may remain. We therefore examine four prominent alternative explanations: monetary tightening, post-pandemic hiring dynamics, the expansion of remote work, and labor-supply shocks. These tests support our interpretation, but they cannot rule out all alternative explanations. In addition, our adoption measure captures deliberate organizational integration of GenAI but may miss informal or “silent” adop-

tion within firms. Finally, because GenAI is still in the early stages of diffusion, longer-run adjustments in training, task allocation, wages, and internal career ladders may either attenuate or amplify the effects documented here.

Even with these caveats, the results suggest that GenAI adoption may be shifting labor demand away from entry-level tasks and narrowing the lower rungs of internal career ladders. Because early-career jobs are central to skill formation and lifetime earnings growth, such changes may have consequences beyond short-run employment. Whether these patterns persist, and whether firms respond by redesigning training, tasks, or career progression, remains an important question for future research.

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